

Stochastic Modeling and AI-Based Calibration for Financial Risk Estimation

1 Introduction

Financial markets exhibit complex and random behavior, making accurate modeling of asset prices a challenging task. Traditional models such as the Black-Scholes Model, and simple statistical models, often fail to capture important empirical features such as time-varying volatility and extreme market events.

The objective of this project is to:

- Model asset price dynamics using Geometric Brownian Motion (GBM) and the Heston Stochastic Volatility model.
 - Estimate financial risk using Value-at-Risk (VaR).
 - Compare simulation outputs with real market data.
 - Improve model accuracy using an optimization-based calibration approach (an AI component).
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2 Stochastic Models

2.1 Geometric Brownian Motion (GBM)

The GBM is defined by the following Stochastic Differential Equation (SDE):

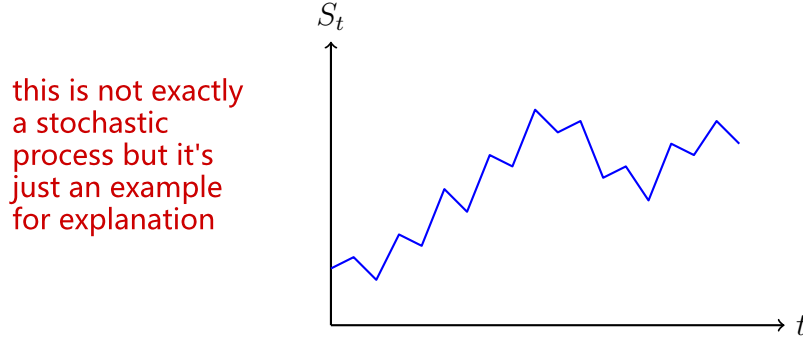
$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

Where:

- S_t : Asset price at time t . It is a stochastic process; it acts like a function of time, but it is not deterministic. Every time we simulate a solution, we get a different function because it is random.
- μ : Drift. This represents the average growth rate. It affects the average trajectory over time ($\mu S_t dt$) and is assumed to be a constant.
- σ : Volatility. This is a constant that dictates how much S_t vibrates or fluctuates.

- W_t : Brownian motion (Wiener process), which is a standard stochastic process.

Interpretation: The GBM model assumes that asset prices evolve with a constant volatility (σ), driven by random fluctuations.



2.2 Heston Model (Stochastic Volatility)

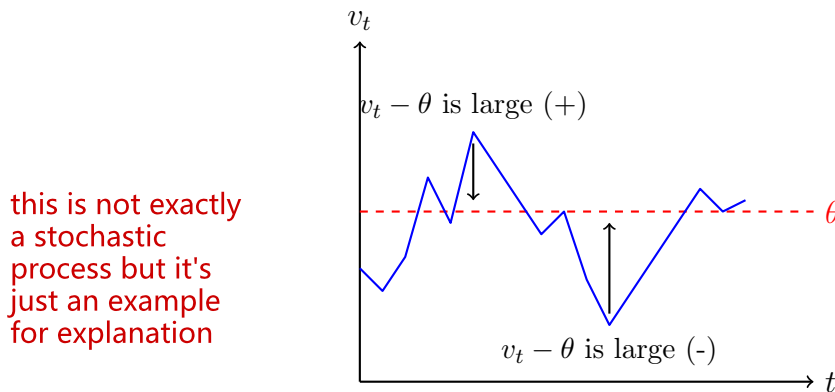
To address the limitations of constant volatility, the Heston model defines the asset price alongside a variance process:

$$\begin{aligned} dS_t &= \mu S_t dt + \sqrt{v_t} S_t dW_t^1 \\ dv_t &= \kappa(\theta - v_t)dt + \xi \sqrt{v_t} dW_t^2 \end{aligned}$$

Where:

- v_t : Variance (which is now time-dependent, rather than a constant).
- κ : Speed of mean-reversion. It controls the speed at which the variance process will revert to its long-term mean.
- θ : Long-term mean of the variance.
- ξ : Volatility of the variance (often called “vol of vol”).
- dW_t^1, dW_t^2 : Two Wiener processes.
- ρ : Correlation of the two Wiener processes, where $cov(dW_t^1, dW_t^2) = \rho dt$. It controls the relationship between the dynamics of the underlying asset price and its volatility. For stocks, this is typically negative.

Interpretation: The Heston model allows volatility to evolve randomly over time while staying anchored to a long-term average, making it far more realistic for financial markets.



3 Discretization (Simulation Form)

To generate paths on a computer, we must divide our time horizon T into small, discrete steps of size Δt using the Euler-Maruyama method.

3.1 GBM Discretization

$$S_{t+\Delta t} = S_t + \mu S_t \Delta t + \sigma S_t \sqrt{\Delta t} Z$$

Here, $Z \sim N(0, 1)$ is a random number drawn from a standard normal distribution.

3.2 Heston Model Discretization

$$v_{t+\Delta t} = v_t + \kappa(\theta - v_t)\Delta t + \xi\sqrt{v_t\Delta t}Z_2$$

$$S_{t+\Delta t} = S_t + \mu S_t \Delta t + \sqrt{v_t} S_t \sqrt{\Delta t} Z_1$$

Where $Z_1, Z_2 \sim N(0, 1)$ are standard normal random variables, and their correlation is $\text{corr}(Z_1, Z_2) = \rho$.

4 Simulation Setup

- **Time horizon:** $T = 1$ year.
 - **Number of steps:** 252 (Standard trading days in a year).
 - **Number of simulations:** 10,000 (To create a robust probability distribution).
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5 Outputs and Risk Measurement

5.1 Return Definition

We calculate the simple return R for each of the 10,000 simulated paths:

$$R = \frac{S_T - S_0}{S_0}$$

5.2 Value-at-Risk (VaR) Explained

Financial Meaning: Value-at-Risk answers a fundamental question for any investor or bank: *“Under normal market conditions, what is the maximum amount of money I can expect to lose over a given time period, with a certain level of confidence?”*

If an investment portfolio has a 1-year 95% VaR of -15%, it means we are 95% confident that the portfolio will not lose more than 15% of its value over the next year. Conversely, there is a 5% probability that the portfolio will experience a loss worse than 15%.

Mathematical Definition and Computation:

Mathematically, for a given confidence level $1 - \alpha$ (where $\alpha \in (0, 1)$ is the significance level), the VaR is defined as the threshold value such that the probability of the loss exceeding this threshold is exactly α :

$$P(R \leq -\text{VaR}_\alpha) = \alpha$$

Since we are using Monte Carlo simulations, we compute this empirically using order statistics. Let N be the total number of simulated paths ($N = 10,000$). We calculate the return for each path, yielding a set of returns $\{R_1, R_2, \dots, R_N\}$. We then sort these returns from the lowest (most severe loss) to the highest (greatest gain):

$$R_{(1)} \leq R_{(2)} \leq \dots \leq R_{(N)}$$

The VaR at confidence level $1 - \alpha$ is simply the absolute value of the return at the α -quantile of this sorted array:

$$\text{VaR}_\alpha = -R_{([N \times \alpha])}$$

Applying this to our simulation:

- **For the 95% VaR** ($\alpha = 0.05$): $N \times \alpha = 10,000 \times 0.05 = 500$. We locate the 500th worst return in our sorted list. $\text{VaR}_{95\%} = -R_{(500)}$.
 - **For the 99% VaR** ($\alpha = 0.01$): $N \times \alpha = 10,000 \times 0.01 = 100$. We locate the 100th worst return in our sorted list. $\text{VaR}_{99\%} = -R_{(100)}$.
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6 Historical Parameter Estimation (Initial Guesses)

Before running optimization algorithms, we must calculate parameters directly from historical data. This acts as our baseline for GBM and the crucial “initial seed” for the Heston model.

6.1 Estimating GBM Parameters

- **Log-returns:** $r_t = \ln\left(\frac{S_{t+1}}{S_t}\right)$
- **Volatility:** $\sigma = \text{std}(r)/\sqrt{\Delta t}$
- **Drift:** $\mu = \text{mean}(r)/\Delta t + \frac{1}{2}\sigma^2$

6.2 Estimating Heston Parameters (Proxy Method)

Because the variance v_t is hidden in real life, we must create a proxy (such as a 30-day rolling historical variance or the VIX index) to estimate the parameters prior to AI calibration.

- **θ (Long-term mean):** The historical average of the variance proxy.

- v_0 (**Initial variance**): The most recent value of the variance proxy.
 - ξ (**Vol of vol**): The standard deviation of the daily changes in the variance proxy.
 - ρ (**Correlation**): The statistical correlation between the daily stock returns and the daily changes in the variance proxy.
 - κ (**Mean reversion speed**): Found by running a linear regression where the dependent variable (Y) is the daily change in variance, and the independent variable (X) is the distance from the mean.
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7 Model Comparison

We will compute and compare the following metrics across the GBM simulation, the historically-estimated Heston simulation, and the real market data:

- Mean of returns
 - Variance of returns
 - Value-at-Risk (VaR) at 95% and 99%
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8 Calibration (AI Component)

Because the historical proxy method is imperfect, we use an optimization algorithm to fine-tune the Heston parameters.

8.1 Objective and Seed Values

Minimize the error over the parameter space $(\kappa, \theta, \xi, \rho, v_0)$. We will feed the historical estimates calculated in Section 6.2 into the optimizer as the starting “seed” values to ensure the algorithm converges efficiently.

8.2 Loss Function

We define a Loss Function (L) that calculates the penalty for how far off our simulation is from the real market data.

$$L = w_1(\text{mean}_{sim} - \text{mean}_{real})^2 + w_2(\text{variance}_{sim} - \text{variance}_{real})^2 + w_3(\text{VaR}_{sim} - \text{VaR}_{real})^2$$

8.3 Interpretation

The AI algorithm will start at our historical guesses, run the simulation, check the loss L , and intelligently adjust the parameters to minimize the difference between simulated outputs and reality.

9 Optimization Strategy

- Use a gradient-free optimization algorithm (such as Nelder-Mead or a Genetic Algorithm).
 - Evaluate the loss function iteratively.
 - Update the parameters until the loss function reaches a minimum threshold, yielding our final, calibrated Heston parameters.
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10 Expected Outputs

Graphs

- Visual plots of simulated price paths comparing GBM and Heston to observe volatility clustering.
- Histograms showing the return distributions (overlaying GBM, Heston, and Real Data).

Quantitative Results & Statistical Comparisons

- VaR (95%, 99%) calculated for each model phase.
 - Historically estimated GBM parameters and Heston parameters.
 - AI-Optimized Heston parameters (Post-calibration).
 - **Error Analysis:** We will compare the statistical information errors (such as the mean and variance) of the GBM Baseline, Heston (Historical), and Heston (AI Calibrated) models directly with the real market data.
 - **Risk Evaluation:** We will analyze the differences in the VaR 95% and VaR 99% across these three modeling phases to determine which model configuration most accurately captures the extreme risk and “fat tails” observed in real-world trading.
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11 Conclusion

This project combines stochastic modeling, statistical estimation, and algorithmic optimization to improve financial risk estimation. While GBM provides a baseline, the Heston model captures realistic market dynamics. By taking historical statistical estimates and refining them through an AI-based calibration loop, this workflow forces the simulated distributions to closely align with real observed market crashes and heavy tails. Accurate financial modeling requires both an appropriate stochastic structure and effective parameter calibration, which this project successfully demonstrates.